

## Submission STAT1-PS1-SUB01

### Question STAT1-PS1-Q01

scratch: show step

1.  $\bar{x} = 33/5 = 6.60$

2.  $s^2 = \sum(x_i - \bar{x})^2 / (5-1) = 18.30$

=> FINAL: mean=6.60; sample variance=18.30 (check units/interpretation)

### Question STAT1-PS1-Q02

scratch: show step

1.  $P(D_1 \cap D_2) = (3/16)((3-1)/(16-1))$

2. =0.0250; the population shrinks after draw 1

=> FINAL: 0.0250 (check units/interpretation)

### Question STAT1-PS1-Q03

scratch: show step

1.  $SE = \sqrt{p(1-p)/n} = \sqrt{0.35(1-0.35)/40} = 0.0754$

2.  $CI = 0.35 \pm 1.96(0.0754) = (0.202, 0.498)$

3. In repeated samples, the procedure captures the population proportion about 95% of the time.

=> FINAL: (0.202, 0.498); repeated-sampling interpretation (check units/interpretation)

### Question STAT1-PS1-Q04

scratch: show step

1. Increase sample size n to reduce standard error.

2. Reduce confidence level or population variability when justified.

3. Lower confidence changes coverage, so precision is not free.

=> FINAL: increase n; lower confidence level or variability; note the coverage trade-off (check units/interpretation)

### Question STAT1-PS1-Q05

scratch: show step

1.  $0.03 < 0.05$ , so reject  $H_0$  under the stated rule.

2. Conclude evidence is inconsistent with  $H_0$ ; do not say the probability  $H_0$  is 3%.

=> FINAL: reject  $H_0$ ; evidence against  $H_0$ , not proof that  $H_0$  is false (check units/interpretation)

### Question STAT1-PS1-Q06

scratch: show step

1.  $\bar{x} = 21/5 = 4.20$

2.  $s^2 = \sum(x_i - \bar{x})^2 / (5-1) = 4.70$

=> FINAL: mean=4.20; sample variance=4.70 (check units/interpretation)